



Experiment No. 7

TITLE: Social Media Analytics Dashboard using Data Visualization Techniques.

AIM: To design and develop a social media analytics dashboard using data visualization techniques to analyze user engagement, sentiment distribution, and trending topics.

THEORY:

1. Introduction to Social Media Analytics Dashboard

Social Media Analytics focuses on extracting meaningful insights from user-generated data on social platforms. Unlike structure-based analytics, which studies relationships between users, this experiment emphasizes data-driven insights using engagement metrics and visualization.

A dashboard provides a visual interface to monitor:

- User engagement (likes, comments, shares)
- Sentiment trends (positive, negative, neutral)
- Trending topics (hashtags)
- High-performing content

Such dashboards are widely used by businesses for marketing strategies, audience analysis, and decision-making.

2. Fundamental Concepts in Social Media Data Analysis

2.1 Data Representation

Social media data is structured into tabular format using DataFrames.

Each row represents a post with attributes such as:

- User
- Post content
- Likes, comments, shares
- Hashtag
- Sentiment

This structured representation helps in efficient analysis.

2.2 Engagement Metrics

Engagement measures how users interact with content.

Engagement Score Formula:

$$\text{Engagement} = \text{Likes} + \text{Comments} + \text{Shares}$$

This metric helps identify:

- Popular posts
- User interaction levels
- Content effectiveness



2.3 Data Simulation

Since real-time API data is not used, synthetic data is generated using tools like Faker.

Generated data includes:

- Random usernames
- Random posts
- Random engagement values
- Random sentiment labels

This simulates real-world social media activity.

3. Core Techniques in Social Media Analytics

3.1 Sentiment Analysis

Sentiment analysis classifies posts into:

- Positive
- Negative
- Neutral

It helps understand public opinion and user mood.

Visualization:

- Pie chart showing sentiment distribution

Business Application:

- Brand perception analysis
- Customer feedback monitoring

3.2 Trending Hashtag Analysis

Hashtags represent topics or trends in social media.

Frequency analysis is used to identify:

- Most used hashtags
- Popular topics

Visualization:

- Bar chart of hashtag counts

Business Application:

- Trend identification
- Campaign optimization

3.3 Engagement Analysis

Analyzes relationship between:

- Likes
- Shares



- Sentiment

Visualization:

- Scatter plot
- Bubble size represents engagement

Business Application:

- Identify high-performing content
- Optimize post strategies

3.4 Viral Content Identification

Posts with highest engagement are considered viral.

Technique:

- Sorting posts by engagement score
- Selecting top-performing posts

Business Application:

- Content strategy improvement
- Influencer collaboration

3.5 Report Generation

A summary report provides key statistics:

- Total posts
- Average likes, comments, shares
- Most trending hashtag

Business Application:

- Quick insights for decision-making
- Performance tracking

4. Applications in Business

Social Media Analytics Dashboard is used in:

- Marketing Analysis: Understand audience behavior
- Brand Monitoring: Track sentiment and reputation
- Content Strategy: Identify what works best
- Trend Analysis: Detect trending topics
- Customer Engagement: Improve interaction strategies

STEPS:

Steps to Implement Social Media Analytics Dashboard Model:

1. Open Google Colab and create a new notebook

```
!pip install pandas plotly faker

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages (5.24.1)
Collecting faker
  Downloading faker-40.11.0-py3-none-any.whl.metadata (16 kB)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.12/dist-packages (from plotly) (9.1.4)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from plotly) (26.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Downloading faker-40.11.0-py3-none-any.whl (2.0 MB)
2.0/2.0 MB 22.1 MB/s eta 0:00:00

Installing collected packages: faker
Successfully installed faker-40.11.0
```

2. Install required libraries (pandas, plotly, faker)

```
[2]
✓ 2s
import pandas as pd
import random
import plotly.express as px
from faker import Faker
```

3. Generate simulated social media dataset

```
fake = Faker()

sentiments = ['Positive', 'Negative', 'Neutral']
hashtags = ['#AI', '#Tech', '#DataScience', '#Python', '#Startup']

data = []

for i in range(200):
    post = {
        "user": fake.user_name(),
        "post_text": fake.sentence(),
        "sentiment": random.choice(sentiments),
        "likes": random.randint(10,500),
        "comments": random.randint(1,100),
        "shares": random.randint(1,50),
        "hashtag": random.choice(hashtags)
    }
    data.append(post)

df = pd.DataFrame(data)

df.head()
```

	user	post_text	sentiment	likes	comments	shares	hashtag
0	tracycameron	Speak bank suffer result prevent.	Neutral	178	100	4	#Tech
1	pbrown	Someone generation study case nor.	Positive	295	71	31	#Startup
2	lowen	Since notice character much else business share.	Neutral	14	29	19	#DataScience
3	eayers	Central live standard water.	Positive	231	11	6	#Tech
4	christopher23	Minute mission number lot or against maintain ...	Neutral	474	39	37	#Python

4. Create engagement metric

```
df["engagement"] = df["likes"] + df["comments"] + df["shares"]

df.head()
```

	user	post_text	sentiment	likes	comments	shares	hashtag	engagement
0	jamesmccullough	Institution beautiful each culture mother.	Positive	376	60	45	#Startup	481
1	elizabeth03	Go purpose want big question color.	Neutral	417	1	45	#Startup	463
2	jsingleton	Success it group popular teacher member.	Neutral	290	11	10	#DataScience	311
3	jparsons	Technology wide learn peace time hot set.	Neutral	311	95	4	#Tech	410
4	james63	Guess fund pay strategy.	Neutral	151	100	27	#Startup	278



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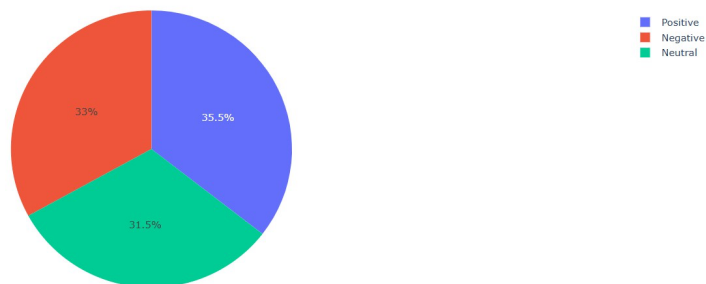
5. Perform sentiment analysis and visualize results

```
sentiment_count = df["sentiment"].value_counts().reset_index()
sentiment_count.columns = ["Sentiment", "Count"]

fig = px.pie(
    sentiment_count,
    values="Count",
    names="Sentiment",
    title="Sentiment Distribution"
)

fig.show()
```

Sentiment Distribution



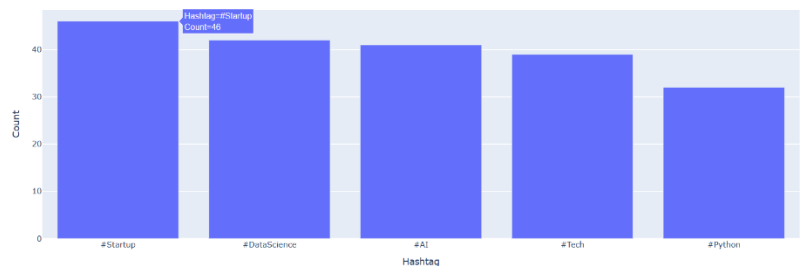
6. Analyze trending hashtags using bar charts

```
hashtag_count = df["hashtag"].value_counts().reset_index()
hashtag_count.columns = ["Hashtag", "Count"]

fig = px.bar(
    hashtag_count,
    x="Hashtag",
    y="Count",
    title="Trending Hashtags"
)

fig.show()
```

Trending Hashtags

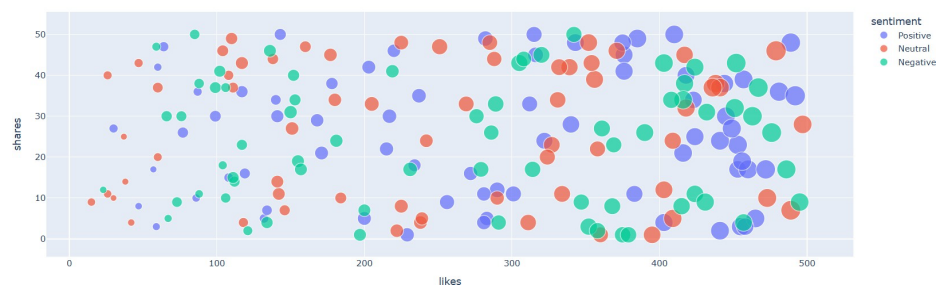


7. Perform engagement analysis using scatter plots

```
fig = px.scatter(
    df,
    x="likes",
    y="shares",
    color="sentiment",
    size="engagement",
    title="Engagement Analysis"
)

fig.show()
```

Engagement Analysis





8. Identify top viral posts

```
top_posts = df.sort_values(by="engagement", ascending=False).head(10)  
top_posts[["user", "post_text", "engagement"]]
```

	user	post_text	engagement
151	mboyd	Technology past evening rich.	626
143	frederickjames	Believe professor us student appear similar.	623
13	williamsbrandon	Maybe simply thousand least number.	596
168	colemanbeth	How wait course of hold.	596
63	smithtracy	Measure career produce move matter he news nice.	587
123	ijames	Make book at year set industry.	585
94	richardburke	Lay success voice use the themselves.	580
24	christopher35	Work song relationship financial see suggest h...	575
163	christinehamilton	Knowledge keep position fund add hundred vote ...	574
149	waltersrussell	Various business court job.	569

9. Generate a summary report

```
report = {  
    "Total Posts": len(df),  
    "Average Likes": df["likes"].mean(),  
    "Average Comments": df["comments"].mean(),  
    "Average Shares": df["shares"].mean(),  
    "Most Trending Hashtag": df["hashtag"].mode()[0]  
}
```

report

```
{'Total Posts': 200,  
 'Average Likes': np.float64(267.3),  
 'Average Comments': np.float64(51.735),  
 'Average Shares': np.float64(25.315),  
 'Most Trending Hashtag': '#Startup'}
```

CONCLUSION:

The experiment demonstrates how a Social Media Analytics Dashboard can be developed using data visualization techniques. It helps in analyzing engagement, identifying trends, and understanding user sentiment. Such dashboards play a crucial role in modern business environments for data-driven decision-making and digital marketing strategies.



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Code:

```
# Install libraries
```

```
!pip install faker plotly
```

```
# Import libraries
```

```
import pandas as pd
```

```
import random
```

```
from faker import Faker
```

```
import plotly.express as px
```

```
fake = Faker()
```

```
# Generate simulated dataset
```

```
data = []
```

```
hashtags = ["#AI", "#DataScience", "#MachineLearning", "#Python", "#BigData"]
```

```
sentiments = ["Positive", "Negative", "Neutral"]
```

```
for i in range(100):
```

```
    likes = random.randint(10,500)
```

```
    comments = random.randint(5,200)
```

```
    shares = random.randint(1,100)
```

```
    data.append({
```

```
        "User": fake.user_name(),
```

```
        "Post": fake.sentence(),
```

```
        "Likes": likes,
```

```
        "Comments": comments,
```

```
        "Shares": shares,
```

```
        "Hashtag": random.choice(hashtags),
```

```
        "Sentiment": random.choice(sentiments)
```

```
    })
```



```
df = pd.DataFrame(data)

# Engagement score
df["Engagement"] = df["Likes"] + df["Comments"] + df["Shares"]

print("Sample Data")
print(df.head())

sentiment_count = df["Sentiment"].value_counts()

fig = px.pie(
    values=sentiment_count.values,
    names=sentiment_count.index,
    title="Sentiment Distribution"
)

fig.show()

hashtag_count = df["Hashtag"].value_counts()

fig = px.bar(
    x=hashtag_count.index,
    y=hashtag_count.values,
    title="Trending Hashtags",
    labels={"x": "Hashtag", "y": "Count"}
)

fig.show()

fig = px.scatter(
```



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```
df,  
x="Likes",  
y="Shares",  
size="Engagement",  
color="Sentiment",  
title="Engagement Analysis"  
)  
  
fig.show()  
  
viral_posts = df.sort_values(by="Engagement", ascending=False).head(5)  
  
print("Top Viral Posts")  
print(viral_posts[["User", "Post", "Engagement"]])  
  
print("Total Posts:", len(df))  
print("Average Likes:", df["Likes"].mean())  
print("Average Comments:", df["Comments"].mean())  
print("Average Shares:", df["Shares"].mean())  
  
print("Most Trending Hashtag:", df["Hashtag"].value_counts().idxmax())
```